

This ensures $\int f d\mu$ is well-defined

#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

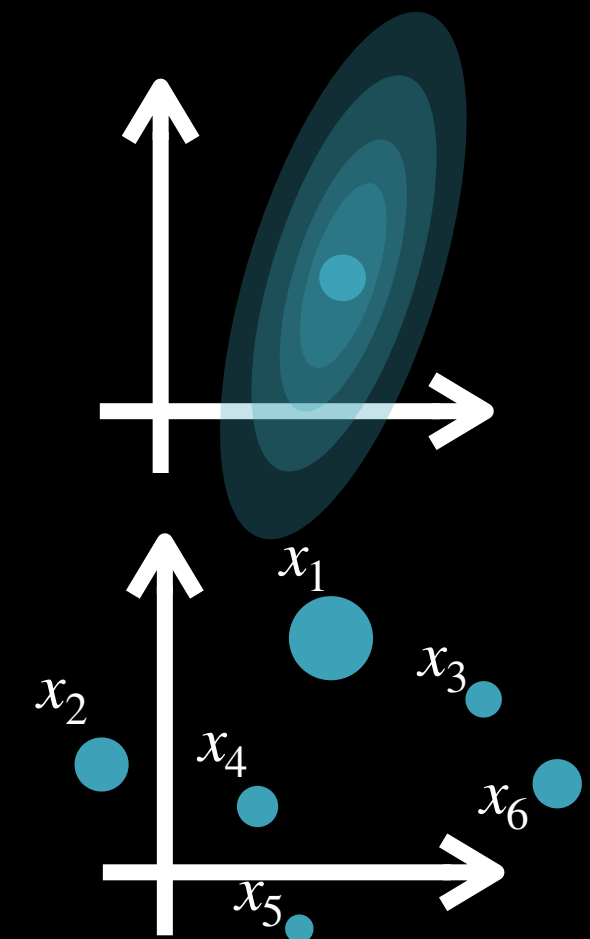


Measure theory in a nutshell

- A function $f: \mathcal{X} \rightarrow \mathbb{R}$ is *measurable* if the preimage of any “nice” set is measurable: For all $B \in \mathcal{B}(\mathbb{R})$, $f^{-1}(B) \in \Sigma$. This ensures $\int f d\mu$ is well-defined
- We integrate measurable functions *with respect to a measure*: $\mathbb{E}_\mu[f] = \int f(x) d\mu(x)$
- Lebesgue measure is the “default” measure on $\mathbb{R}^n$. It assigns mass equal to volume (e.g., length, area, volume, hyper-volume).
- Special cases of measures and integration against them:

Continuous: $\mu(x) = \rho(x)dx \implies \int f d\mu = \int f(x)\rho(x)dx$

Discrete: $\mu = \sum_i p_i \delta_{x_i} \implies \int f d\mu = \sum_i p_i f(x_i)$



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